



Robots fight drought

Researchers are using robotic platforms to find drought resistant strains of crop. <http://dgit.in/RoboCrop>



Trees=global warming?

Researchers find that tree trunks do emit methane. But do they cause global warming? <http://dgit.in/warmtrees>

The hope and despair of Artificial Wombs

Growing humans artificially is a futuristic dream for some, and a macabre dystopia devoid of the essence of life for others. How close are we to building an artificial womb?

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It is often a source of wonder when a technology once deemed as 'wishful' thinking, the stuff of legends or part of science fiction lore is actually realised. Think self-lacing shoes that Nike recently unveiled or how we now have the Roomba, a robotic vacuum cleaner! But, the possibilities and realisations aren't always half as harmless or cute.



Is this how life ought to begin? We say why not.

A rule, now popularly known as the "fourteen-day rule" was first proposed in 1979 and has since then been incorporated into the law books and guidelines of numerous countries. Unlike building computing devices, holograms or robots, growing human embryos in-vitro has deep moral and ethical implications. Keeping that in mind, the fourteen day rule forbids experiments on human embryos to

carry on beyond fourteen days, widely recognised as the time in which an embryo develops a semblance of individuality with the start of a head-tail axis. Back then, people who formulated this law probably also believed that technology might never advance to a point to which we can produce a fully functional human baby without natural womb support – the rule will never have to be amended or challenged even. Despite rapid advances in medicine, biology and engineering, raising a foetus in a lab environment, ectogenesis, has remained a pipe dream. In this feature we give you a lowdown on why.

The Intricacies

Let's start with how incredibly sophisticated and complicated the developmental process of a human foetus is. At the very onset of the above developmental

process is a stage called implantation. Post implantation is when there is self organization and the embryo becomes a collection of cells. This stage is critical because it is during this time, when many of the developmental disorders set in. The week around implantation - when the foetus is essentially attached to the mother's uterus, is one of the most important and yet poorly understood phases of pregnancy. Sci-

entists have been unable to put a tab on intricate cellular interactions, primarily because it's such a difficult system to get results out of. Growing embryo's in a petri dish and gaining the ability to culture the embryo artificially beyond implantation is something that was possible only for simpler organisms like mice until very recently. Last year, however, two independent research groups reported successful culture of human embryo's for thirteen days. By providing the nascent embryo with a specific cocktail of growth supporting nutrients, the researchers found that the embryos were able to self organise. Their results provide irrefutable evidence that a human embryo can develop beyond the implantation stage without maternal input of any sorts, in-vitro - in completely artificial conditions. Besides being a foundation setting study it has paved the way for a large number of offshoot research to study birth defects, disease and fundamentals of developmental biology. Another important point to note here is that the embryo culture continued for only thirteen days because the researchers had to pull the plug in adherence to the fourteen day rule. However, ethical and legal issues aside, we are now in a position to culture embryos artificially for a minimum of two weeks. The next step - ensuring survival of the embryo once it begins to form structures.